

9th World Mathematics Team Championship 2018

Advanced Level Round 1 Numerical Answers

1	2	3	4	5	6	7	8
C) 2	A) -5	D) $\frac{\sqrt{15}}{8}$	D) 2520	E) 15 cm	C) 576	D) 8	B) 2

9	10	11	12	13	14	15
E) 5	B)	B) 5	A) 42	B) [3,4]	A) $\frac{4a}{3(2-a)}$	D) $-\frac{3}{2}$

Advanced Level Round 1 Detailed Solutions

1. $\sqrt{9} + \sqrt{(3 - \sqrt{11})^2} + \sqrt[3]{(2 - \sqrt{11})^3} = 3 + \sqrt{11} - 3 + 2 - \sqrt{11} = 2.$

2. Note that $2^{x-1} \times 5^{x-1} = 10^{x-1}$ and $0.1 \times 10^{2x+5} = 10^{2x+4}$. It follows that $x - 1 = 2x + 4$, or $x = -5$.

3. From $\sin \alpha = \pm \sqrt{1 - \cos^2 \alpha} = \pm \frac{\sqrt{15}}{4}$ and $\sin \alpha < 0$ we obtain $\sin \alpha = -\frac{\sqrt{15}}{4}$.

Thus, $\sin 2\alpha = 2 \sin \alpha \times \cos \alpha = \frac{\sqrt{15}}{8}.$

4. There are 10 choices for the leader, 9 for the deputy leader and $\binom{8}{2} = 28$ for the two members. For the whole group there are $10 \times 9 \times 28 = 2520$ choices.

5. Denote the length of the unknown side by x . The altitude to the side of length x cm is less than 9 cm and therefore it is the altitude of length 6 cm. Then $S = 9 \times 10 = 6 \times x$ giving $x = 15$ cm.

6. All arrangements of the six girls are $6! = 720$. The arrangements for which the three Chinese girls are sitting together next to each other are $4! \times 3! = 144$. Therefore the answer is $720 - 144 = 576$.

7. Since $\sqrt{3 - \sqrt{5}} = \sqrt{\frac{6 - 2\sqrt{5}}{2}} = \frac{\sqrt{5} - 1}{\sqrt{2}}$ we have

$$(3 + \sqrt{5})(\sqrt{10} - \sqrt{2})\sqrt{3 - \sqrt{5}} = (3 + \sqrt{5})(\sqrt{5} - 1)^2 = 2(3 + \sqrt{5})(3 - \sqrt{5}) = 8.$$

8. For $u = x + y$ and $v = xy$ the system becomes $\begin{cases} u + v = 7 \\ u \times v = 12 \end{cases}$ with solutions $u = 3, v = 4$ and $u = 4, v = 3$. The first pair doesn't lead to solutions for x and y and the second one gives two solutions. /

9. Since the number of 2-digit numbers having only odd digits is $5 \times 5 = 25$ we have that each odd digit appears as unit digit 25 times. Thus the unit digit of the sum of all 3 digit numbers having only odd digits equals the unit digit of $(1 + 3 + 5 + 7 + 9) \times 25 = 25 \times 25$, i.e. it is 5.

10. Since the range is $\left[-\frac{1}{8}, \infty\right)$ we have that $a > 0$. Then the minimum value of $y = ax^2 + c$ equals c , thus the answer is B).

11. Using that $MA^2 + MC^2 = MB^2 + MD^2$ we obtain $29 = 4 + MD^2$ giving $MD = 5$

12. Since $1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ we have

$$(1 + i\sqrt{3})^{21} = \left(2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\right)^{21} = 2^{21}(\cos 7\pi + i\sin 7\pi) = -2^{21}.$$

Thus $a^2 = 2^{42}$ and $\log_2 a^2 = \log_2 2^{42} = 42$.

13. Since $(x^2 - 5x + 4) = (x - 1)(x - 4)$ and $x \geq 3$ the solution is $[3, 4]$.

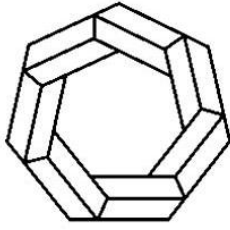
$$14. \log_2 \sqrt[3]{625} = \frac{1}{3} \log_2 625 = \frac{4}{3} \log_2 5$$

$$= \frac{4}{3} \times \frac{\log_{10} 5}{\log_{10} 2} = \frac{4}{3} \times \frac{a}{2 \log_{10} 2}$$

$$= \frac{4}{3} \times \frac{a}{\log_{10} 4} = \frac{4}{3} \times \frac{a}{\log_{10} \frac{100}{25}} = \frac{4}{3} \times \frac{a}{2 - a}.$$

15. By inspection we find that the solution is $-\frac{3}{2}$. It is also possible to use that

$$\frac{x}{x^2 + 2} \sqrt{1 + \frac{x^4 + 4}{4x^2}} = \frac{x}{x^2 + 2} \sqrt{\frac{(x^2 + 2)^2}{(2x)^2}} = \frac{x}{x^2 + 2} \frac{x^2 + 2}{2|x|} = \frac{x}{2|x|} = \begin{cases} \frac{1}{2} & \text{for } x > 0 \\ -\frac{1}{2} & \text{for } x < 0 \end{cases}.$$



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Advanced Level Round 2 Numerical Answers

1	2	3	4	5	6	7	8
35	1	3	2	39	3510	51	26

Advanced Level Round 2 Detailed Solutions

1. Let $n = 2^\alpha \times 3^\beta \times m$ where $\alpha \geq 0$ and $\beta \geq 0$. Then
 $d(2n) = (\alpha + 2)(\beta + 1)d(m) = 28$ and $d(3n) = (\alpha + 1)(\beta + 2)d(m) = 30$. Since $\gcd(28, 30) = 2$ we have $d(m) = 1$ or $d(m) = 2$.
 If $d(m) = 2$ we obtain $(\alpha + 2)(\beta + 1) = 14$ and $(\alpha + 1)(\beta + 2) = 15$ with no solution.
 If $d(m) = 1$ we obtain $(\alpha + 2)(\beta + 1) = 28$ and $(\alpha + 1)(\beta + 2) = 30$ with $\alpha = 5$ and $\beta = 3$. Therefore $d(6n) = (\alpha + 2)(\beta + 2) = 35$.

2. Since $\min \{x^2 - 2x + 5\} = 4$ (the minimum is achieved for $x = 1$) we have

$$\max \left\{ \log_{\frac{1}{2}} (x^2 - 2x + 5) \right\} = -2 \text{ and } \min \left\{ \left(\frac{1}{3} \right)^{\log_{\frac{1}{2}} (x^2 - 2x + 5)} \right\} = 9.$$

Also $\max \{9 - |x - 1|\} = 9$ for $x = 1$. Therefore $x = 1$.

3. Since $\angle BPM = \angle BCM = 90^\circ$ the quadrilateral $PBCM$ is cyclic. Then $\angle PCB = \angle PBM = \angle PCM = 45^\circ$. For $\varphi = \angle CBM$ the sine theorem for triangle PBC implies:

$$\frac{PC}{BC} = \frac{\sin \angle PBC}{\sin \angle BPC} = \frac{\sin(\varphi + 45^\circ)}{\sin(90^\circ - \varphi)} = \frac{\sin(\varphi + 45^\circ)}{\cos(\varphi)} = \frac{\sqrt{2}}{2} \tan \varphi + \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4} + \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{4}$$

For $BC = 1$ we have $PC = \frac{3\sqrt{2}}{4}$ and therefore $\frac{PC}{PA} = \frac{\frac{3\sqrt{2}}{4}}{\sqrt{2} - \frac{3\sqrt{2}}{4}} = 3$.

4. Denote $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$. The $x^2 = 2 + x$ with roots $x = 2$ and $x = -1$. Since $x > 0$ we obtain $x = 2$.

5. Since $AC \perp BD$ we have that $AB^2 + CD^2 = BC^2 + AD^2$. Therefore $AD^2 = AB^2 + CD^2 - BC^2 = 60^2 + 25^2 - 52^2 = 39^2$ and thus $AD = 39$ cm.

The diagonals AC and BD of quadrilateral $ABCD$ are perpendicular. If $AB = 60$ cm, $BC = 52$ cm and $CD = 25$ cm find AD .

6. Denote the number by $\overline{a_1 a_2 a_3 a_4 a_5}$. The number of solutions of

$a_1 + a_2 + a_3 + a_4 + a_5 = 22$ for $a_i > 0$ equals to $\binom{21}{4} = 5985$. From this number we have to subtract the number of solutions having $a_i > 9$ for some $i = 1, 2, 3, 4, 5$ (it is not possible to have $a_i > 9$ and $a_j > 9$ because then $a_1 + a_2 + a_3 + a_4 + a_5 > 22$).

Replacing a_i by $a_i - 9$ leads to $a_1 + a_2 + a_3 + a_4 + a_5 = 13$ which admits $\binom{12}{4} = 495$

The answer is $5985 - 5 \times 495 = 3510$.

7. Denote by a, b, c, d and e the number of problems solved from day 1 to 5; 6 to 11, 12 to 14, 15 to 25 and 26 to 31, respectively. The condition of the problem implies

$a + b + c + d = n, a + b + c = 70, b + c + d = 80$ and $c + d + e = 60$. Therefore

$n + c = (a + b + c) + (c + d + e) = 130$, thus $n = 130 - c$. Since $b + c \leq 70$ it follows that $d \geq 10$ and $c \leq 50$. It is easy to see that any $c = 0, 1, \dots, 50$ gives a solution.

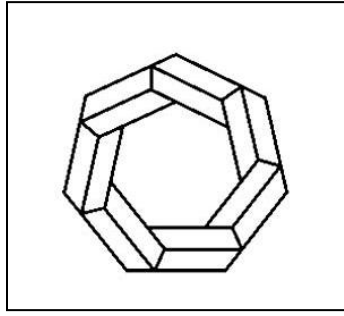
The answer is 51.

8. The balls can be placed on the circle by $\binom{10}{5}$ ways. Consider an arrangement

that doesn't change after a rotation of $36k$ degrees where $k, (1 \leq k \leq 9)$ is a positive integer. Then 10 is a factor of $5k$ implying that k is even. It is easy to see that this is possible only for alternating colors (there are two such arrangements). Then all

remaining $\binom{10}{5} - 2$ arrangements are grouped by 10's. The answer is

$$\frac{\binom{10}{5} - 2}{10} + 1 = 26.$$



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Advanced Relay Detailed Solutions

R1-A With no lines we have one region. The first line adds one more region; the second line adds 2 more regions and so on the n -th line adds n more regions. Therefore the total number of regions equals $1+1+2+\dots+20=1+\frac{20\times 21}{2}=211$.

Answer: 211

$T = 211$

R1-B If $b - a > 2$ we may increase a by 1 and decrease b by 1 having $a + 1 < b - 1$. Then

$$ab + bc + ca = ab + c(T - c) < (a + 1)(b - 1) + c(211 - c)$$

and the value increases. We conclude that $b - a \leq 2$ and the same arguments show that $c - b \leq 2$.

If $b - a = 2$ and $c - b = 2$ then $a + 1$, b and $c - 1$ increase the value of the expression. Therefore:

1. $b - a = 1$ and $c - b = 1$ giving $3b = 211$ with no solution in positive integers;
2. $b - a = 1$ and $c - b = 2$ giving $a = 69$, $b = 70$, $c = 72$
3. $b - a = 2$ and $c - b = 1$ giving $3b = 212$ with no solution in positive integers.

Answer: 70

R2-A We can choose the two positions in which a nice number coincides with 12345 in 10 ways. After that we choose the two positions in which a nice number coincides with 23456 in 3 ways. The remaining digit of a nice number can be chosen in 7 ways. All nice numbers are $10 \times 3 \times 7 = 210$.

Answer: 210

$T = 210$

R2-B Let M be the midpoint of CD . Then $CM = 105$ cm, $OC = 350$ cm and from triangle OCM we have $OM = OM = 35\sqrt{91}$ cm. If x is the distance from C to OD we have $x \times 350 = 210 \times 35\sqrt{91}$ giving $x = 21\sqrt{91}$.

Answer: $x = 21\sqrt{91}$ cm

R3-A Since $\frac{2}{a_n} = \frac{1}{a_{n-1}} + \frac{1}{a_{n+1}}$ the reciprocal values form an arithmetic sequence

with first term $\frac{1}{3}$ and common difference $\frac{1}{6}$. Therefore $\frac{1}{a_{2018}} = \frac{1}{3} + 2017 \cdot \frac{1}{6} = \frac{673}{2}$.

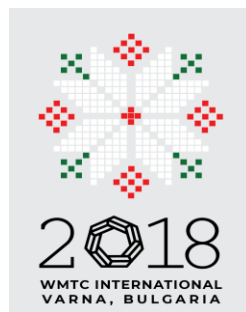
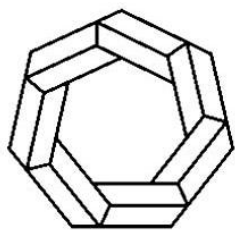
Answer: 673

$T = 673$

R3-B We can delete three letters abc or aaa or bbb or ccc . We are left with a , aa , ab , aab or $aabb$. Only in aab it is not possible to achieve divisibility by 3. So, good are only words of length not divisible by 3.

There are 448 numbers divisible by 3 (from 3×225 to 3×672) in the interval $[673, 2018]$. All numbers are $2018 - 672 = 1346$. Not divisible by 3 are $1346 - 448 = 898$.

Answer: 898



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Advanced Team Detailed Solutions

1. Since the leading coefficient is 1 the root is an integer number. Further, it follows from $a = -x^2 + 2x - 4 - \frac{7}{x+2}$ that $x + 2$ divides 7, i.e. $x = -1, -3, 5$ and -9 . The corresponding values of a are $-14, -12, -20$ and -102 . The least value of a is -102 and then $a^2 = 10404$.

Answer: 10404

2. If x is a root then $-x$ is also a root. To have 5 roots we need a root $x = 0$. Then $b = -a^2$ or $b = a^2 + 2$. In the first case the equation has three roots 0 and $\pm\sqrt{2}$. In the second case there are five distinct roots: 0, $\pm\sqrt{2}$ and $\pm\sqrt{2a^2 + 4}$.

Now the minimum value of $2a + |b| + 49 = a^2 + 2a + 51$ is 50 and is attained for $a = -1$.

Answer: 50

3. Since $(1 \pm \sqrt{2})^3 = 7 \pm 5\sqrt{2}$ we have $a = \sqrt[3]{7} + \sqrt{2} + 1$ and $b = \sqrt[3]{7} + \sqrt{2} - 1$.

Now $a - b = 2$ and after substitution we obtain the answer 7.

Answer: 7

4. The centers of the four balls are vertices of regular pyramid with edge length equal to the diameter $d = 40(3 - \sqrt{6})$ of the ball. The altitude of such a pyramid is $\frac{d\sqrt{6}}{3}$.

The distance from the ground to the top equals

$$\frac{d\sqrt{6}}{3} + d = 40(3 - \sqrt{6}) \left(\frac{\sqrt{6} + 3}{3} \right) = 40 \text{ cm.}$$

Answer: 40 cm

5. Let $1 \leq a_1 < a_2 < \dots < a_k$ be the elements of A . By setting $x = a_1$ and $y = a_2$ we obtain $\frac{1}{a_2} \leq \frac{1}{a_1} - \frac{1}{25} \leq \frac{24}{25}$, i.e. $a_2 \geq 2$. Analogously $a_3 \geq 3$, $a_4 \geq 4$, $a_5 \geq 5$, $a_6 \geq 7$, $a_7 \geq 10$, $a_8 \geq 17$, $a_9 \geq 54$ and $a_{10} < 0$, a contradiction. The set $\{1, 2, 3, 4, 5, 7, 10, 17, 54\}$ has the required property.

Answer: 9

6. Each of the first $5! = 120$ numbers (having altogether 1200 digits) begins with 10234. The last number is 1023498765. Thus, the 1198-th digit equals 7.

Answer: 7

7. For $x = 0$ the inequalities become $b \geq 1 \geq a$. Further, it follows from $ax^2 + bx + 1 \geq bx^2 + x + a$ that $a \geq b$. Finally, $a = b = 1$ and there is only one pair.

Answer: 1

8. Easy induction shows that for the set $\{1, 2, 3, \dots, n, n + 6\}$ we have $S = (n + 1)! - 1$. Therefore $S + 1 = 100!$ and the largest prime divisor of $100!$ is 97.

Answer: 97

9. All considered triples have equal products. It is easy to see that if $1 = d_1 < d_2 < \dots < d_{27} = 900$ are all divisors of 900 then the desired number equals $\tau(d_1) + \tau(d_2) + \dots + \tau(d_{27})$ where $\tau(n)$ is the number of divisors of n . Since $900 = 2^2 \cdot 3^2 \cdot 5^2$ we have $\tau(d_1) + \tau(d_2) + \dots + \tau(d_{27}) = (1 + 2 + 3)^3 = 216$.

Answer: 216

$T = 97$

10. If D is the symmetric point of C with respect to M then $\angle NMK = \angle QDP$.

If $\angle BAC = \alpha$, $\angle ABC = \beta$ and $\angle ACB = \gamma = T^\circ$ then $\angle ADP = \frac{180^\circ - \beta}{2}$ and $\angle BDQ = \frac{180^\circ - \alpha}{2}$. It follows that $\angle QDP = \angle ADP + \angle BDQ - \angle ADB = \frac{180^\circ - \gamma}{2}$, i.e. $2\angle NMK = 83^\circ$.

Answer: 83

$T = 7$

11. $3\binom{9}{4} = 378$. In general the answer is $3\binom{T+2}{4}$

Answer: 378

$T = 1$

12. The equality is equivalent to $(x - 7)(y + 7) = -41$ and the greatest value is attained for $x = -34$ and $y = -6$.

Answer: 204

$T = 50$

13. A person climbs a moving escalator. From the moment he steps on the escalator to the moment he reaches the end of the escalator he climbs T steps. When climbing two times faster from the moment he steps on the escalator to the moment he reaches the end of the escalator he climbs $T + 10$ steps. Find the number of steps the escalator in rest has.

Solution: $\frac{50.60}{2.50 - 60} = 75$ In general the answer is $\frac{pq}{2p - q}$ where p and q are the given number of steps.

Answer: 75

$$T = 9, S = 7$$

14. Denote the tangent point of k_1 and k_2 by I . Since $\angle IAC = \frac{IC}{2} = \frac{IA}{2} = \angle IAB$ we have that IA is angular bisector of angle BAC . The same is true for IC , ID and IB and corresponding angles. Therefore $ACDB$ is circumscribed, giving $AC + BD = 2AB$. But $AB = 2\sqrt{r_1 r_2} = 6\sqrt{7}$ (it follows from $O_1 O_2 BA$) and thus $(AC + BD)^2 = 4AB^2 = 1008$.

Answer: 1008